2/4/2026 Scientific Literature

- Comments on
  - Quiz: make sure to understand foundational material
  - Exercise 3
    - Confirm MOR:Gα instability
    - Carry over to Exercise 4
- Biological target presentation and report
- Searching the scientific literature
  - Biochemistry-related databases
  - Search strategies
  - Accessing articles
  - Zotero
- Library and Resources and Citation Management (Guest workshop by Sean Murphy)

Biochemistry-related databases

- Scientific Literature
  - Google/Google Scholar
  - Web of Science
  - PubMed
  - Others: https://guides.library.iit.edu/c.php?g=474723&p=3248565

- GPCR Database: https://gpcrdb.org/
  - "GPCRdb, GproteinDb, ArrestinDb and Biased Signaling Atlas cover related aspects of signal transduction while serving dedicated research communities and use cases."
- PubChem: https://pubchem.ncbi.nlm.nih.gov/
  - Includes biological activity of many chemical entities
  - Chemicals often identified by SMILES strings

Search strategies

- Consider alternative names, e.g. coronavirus main protease (MPro) and 3c-like (3cl) protease and nonstructural protein 5 (NSP5)
- Proteins from different species are usually similar
- Look at citations
  - cited by an article of interested
  - that cite the article of interest
- A research group will often work on similar proteins

Accessing articles

- Free
  - Open access - article is free online. Usually requires larger author payment
  - Recent articles on NIH-funded research is freely available online at PubMed Central (PMC)
  - Preprint servers, e.g. bioarXiv - free, but may not be peer reviewed
  - Sometimes available from authors on their websites or upon request
- Paywall
  - IIT subscribes to a selection of scientific journals, which can be accessed
    - in the library
    - online from campus
    - via a proxy server or VPN
  - Interlibrary loan (MyILL) can be used to request nearly any article, but requires a lead time

Zotero

- Open source and free
- Available online and as an application
- Automatically organizes manuscript files & notes
- Allows sharing with others via group libraries
- Creates bibliographies
  - linked with MS Word and Google Docs
  - exportable to BibTeX

Library Resources and Citation Management

CHEM 456 - Computational Biochem

Sean Murphy | he/him/his
Head of Systems and Open Infrastructure – Galvin Library
Liaison Librarian - Chemistry, Biology, Physics, Food Science
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How do *you* find information?

Learning Objectives

- Finding Full-Text Articles
- Database Searching
- Tracking the Literature
- Evaluating and Citing Sources
- Library Resources

Finding Sources

Popular Sources vs Scholarly Sources

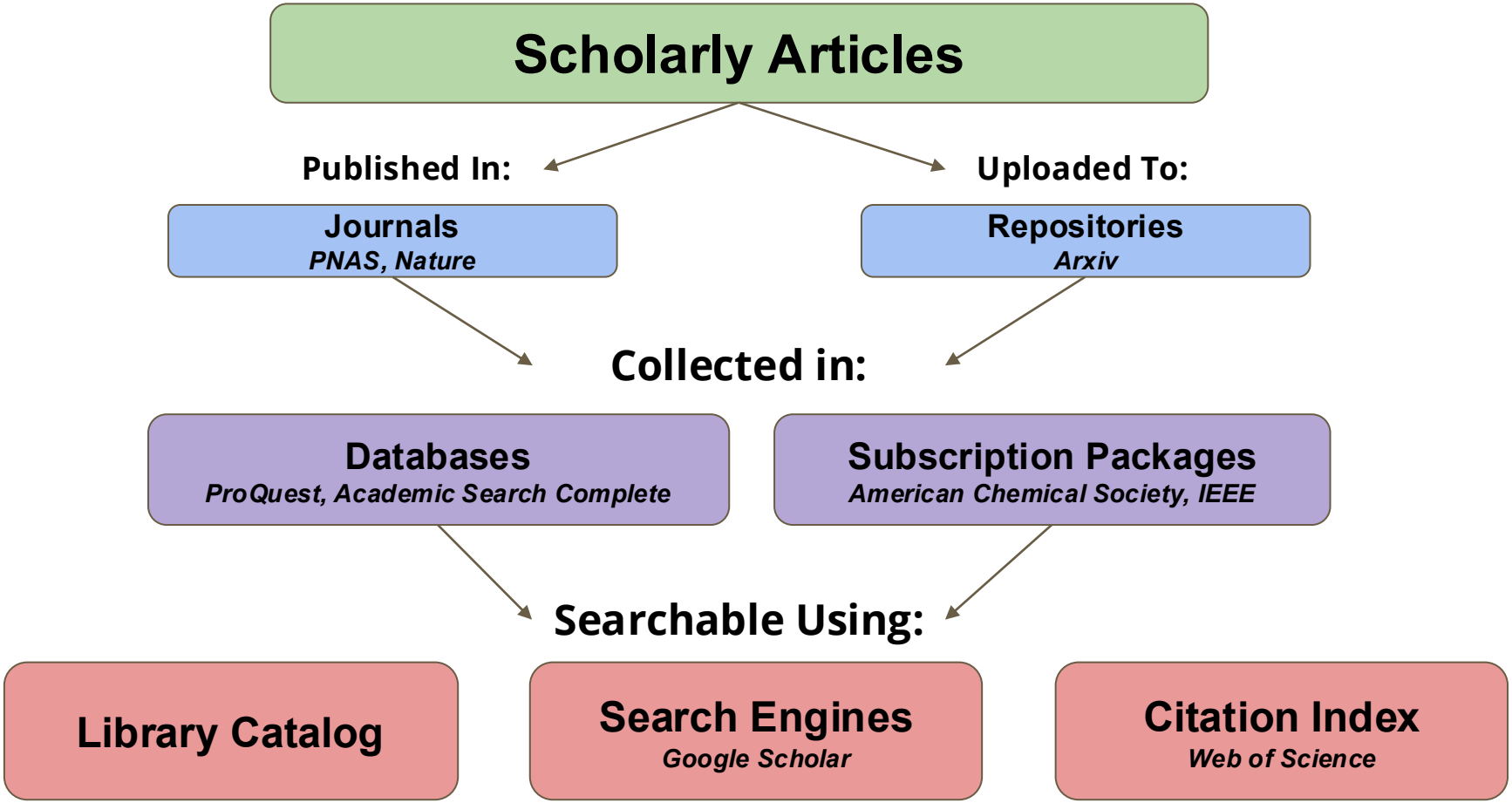
Popular Sources are written for a general audience

They may be intended to inform, entertain, or persuade

Scholarly Sources are written for an academic audience

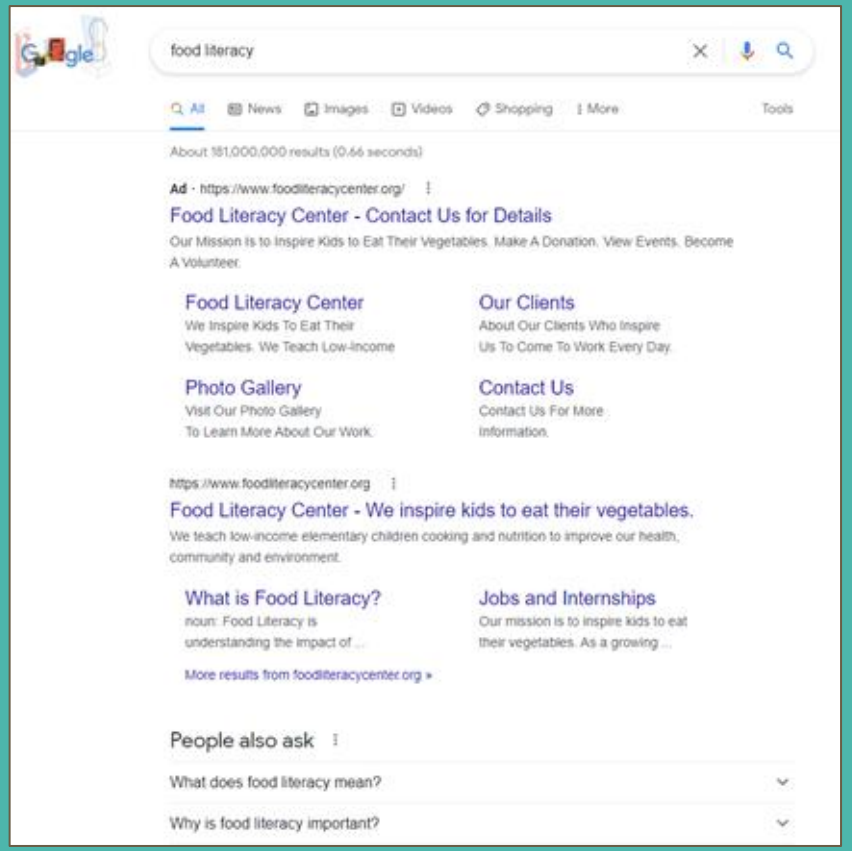
They are intended to contribute to a disciplinary discourse

Peer Review


Scholarly Articles
Published In:
Uploaded To:
Journals
PNAS, Nature
Repositories
Arxiv
Collected in:
Databases
ProQuest, Academic Search Complete
Subscription Packages
American Chemical Society, IEEE
Searchable Using:
Library Catalog
Search Engines
Google Scholar
Citation Index
Web of Science

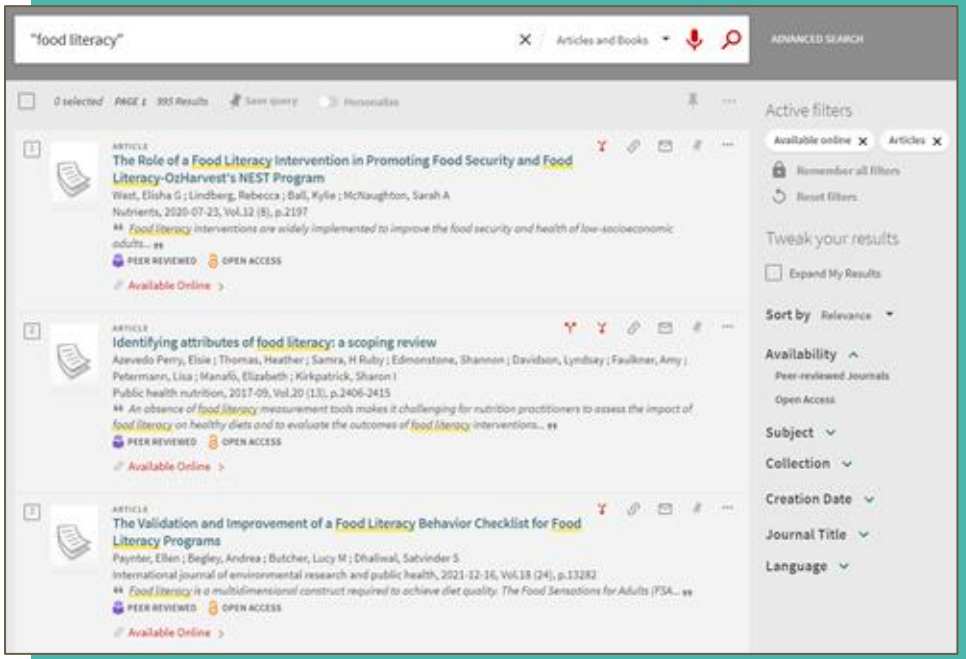
Searching Using Google

Algorithms and “Natural Language Processing”



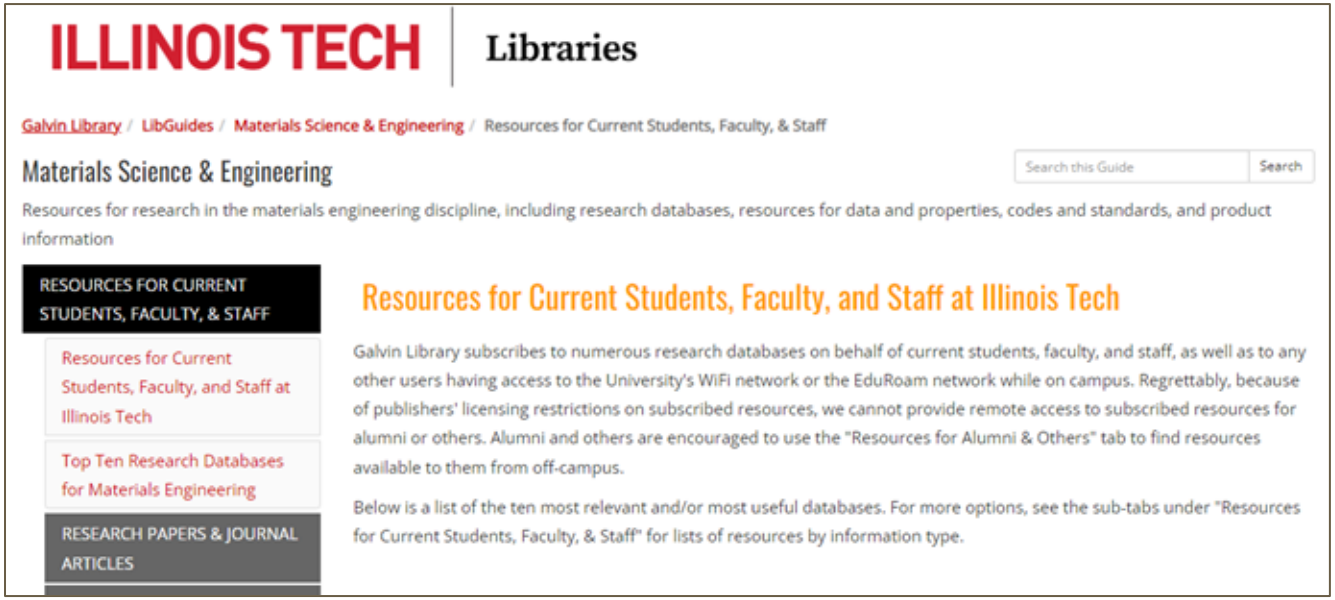
Catalogs/Databases

Keywords and Filters



LibGuides

- Discipline-specific, curated guides
- Compiled by Librarians
- Databases
- Patent and Product Information
- Tips and Tricks
- Research Help



Chemistry

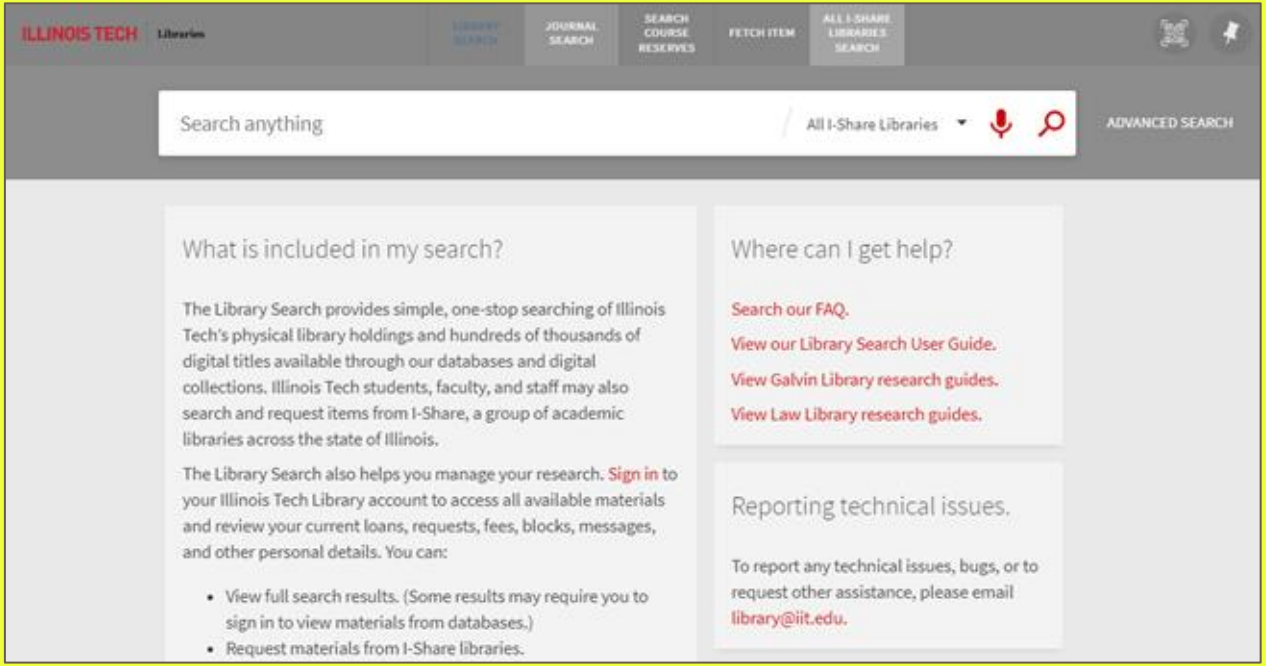
Chemical Engineering

Research Tips

library.iit.edu

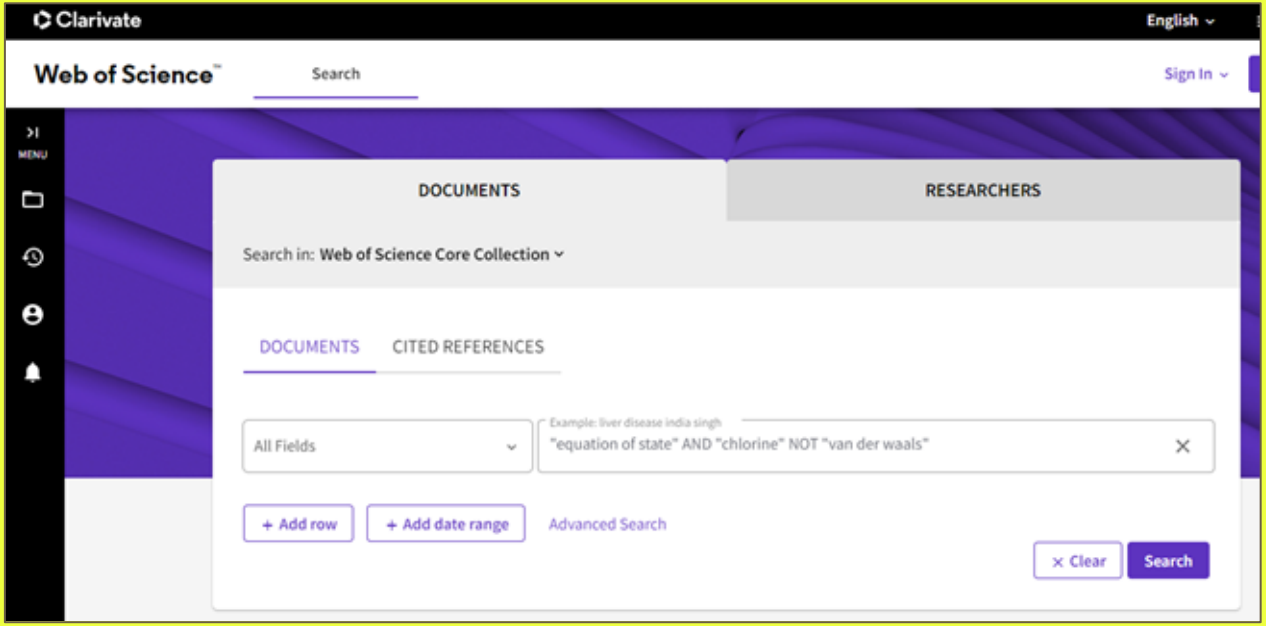
Library Catalog

- **Sign in with HawkID**
- **Searches all library subscriptions**
- **Will show articles, books, databases, etc. in search results**
- **Great starting point**



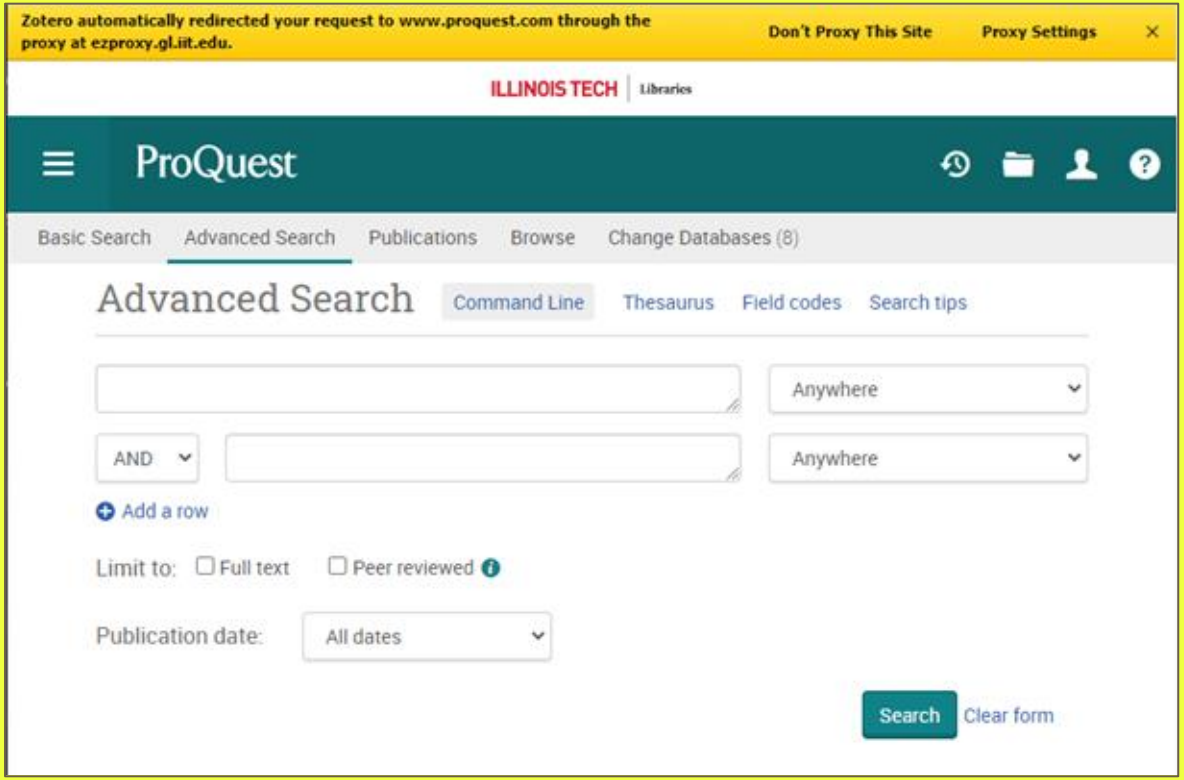
Web of Science

- Citation Index
- Full Text not necessarily available
- Powerful citation tools



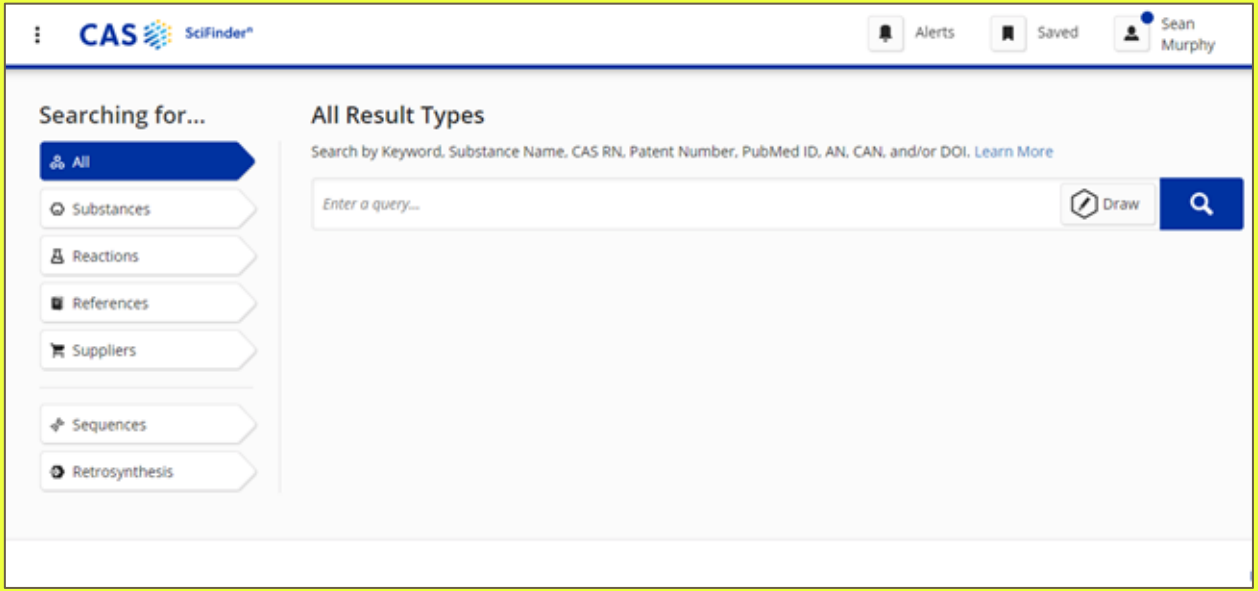
Proquest SciTech

- **Boolean Searching**
- **Limits and Filters**
- **Full Text**
- **“Find Full Text” - Library Catalog**



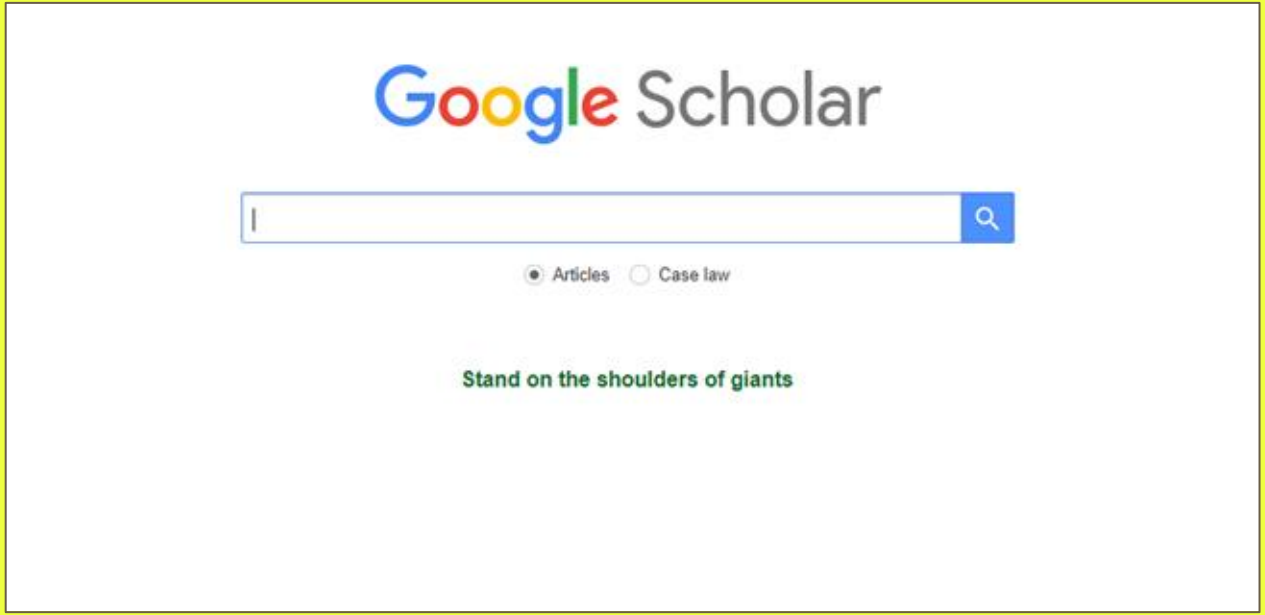
SciFinder

- Requires Additional Signup
- Chemical Structure and Properties Database
- Reference citations for property data



Google Scholar

- **Broad scope, search-engine style interface**
- **Can be linked with IIT holdings**
- **Good for when you know exactly what you're looking for**



Citing Sources

Why Cite?

Give Credit

Demonstrate a scholarly discourse

Support your claims with evidence

Direct readers to original sources

Avoid plagiarism

Plagiarism

Using someone else’s words, ideas, or work as your own without giving due credit to the original author

How do you give credit?

In-Text Citations

Attributing a sentence, quote, or idea to a particular reference within the body of your text

Full Reference

A citation that’s formatted and included at the end of your paper in a bibliography or works cited section

Illinois Tech Code of Academic Honesty

“The misrepresentation of any work submitted for credit or otherwise as the product of a student’s sole independent effort, such as using the ideas of others without attribution and other forms of plagiarism[...],” Illinois Tech Code of Academic Honesty, section I.A

Penalties for violating the Code of Academic Honesty can include:

- Grade reductions
- Expulsion from a course or the university

When in doubt.....

Just Cite!

Citation Styles

Perdue OWL - Online Writing Lab

APA (American Psychological Association) - Sciences

MLA (Modern Language Association) - Humanities

Chicago Manual of Style

In-text citation

Examples from a scholarly article:

Chronic pain and functional limitation have been linked to psychological conditions such as anxiety and depression in a number of studies (Lenze et al., 2001). A form of functional limitation that affects an estimated two percent of adults is referred to as "frozen shoulder" (Dias, Cutts, & Masoud, 2005). Frozen shoulder is charac-

Church, D., & Nelms, J. (2016). Pain, range of motion, and psychological symptoms in a population with frozen shoulder: A randomized controlled dismantling study of clinical EFT (emotional freedom techniques). *Archives of Scientific Psychology, 4*(1), 38-48. doi:10.1037/arc0000028

Scalabrin, G., Bettio, L., Marchi, P., & Stringari, P. (2007). A fundamental equation of state for Sulfur Hexafluoride (SF6) in extended equation of state format. *Journal of Physical and Chemical Reference Data, 36*(2), 617–662. https://doi.org/10.1063/1.2716004

Bibliography

Located at the end of papers

How can readers find the cited resource?

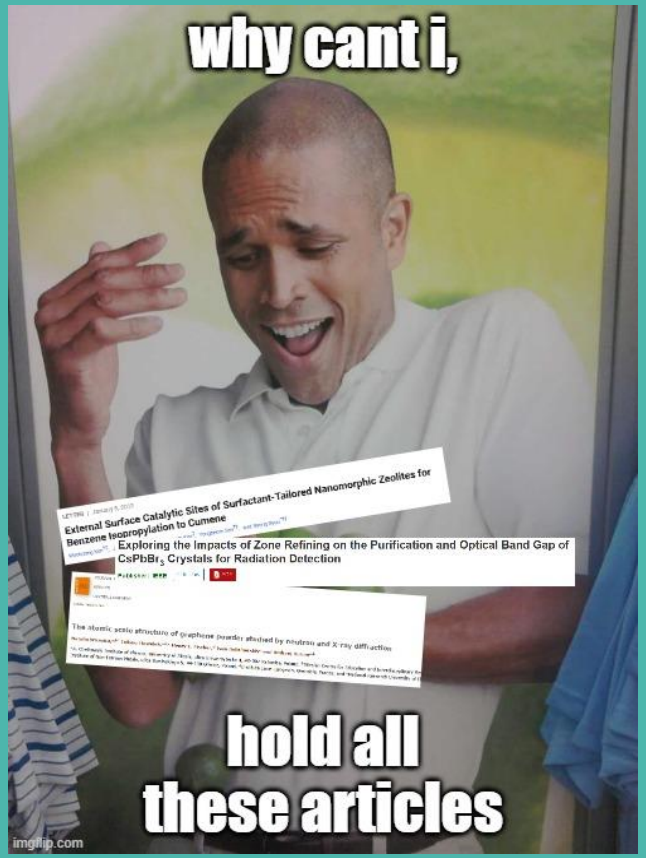
“Bibliographic Information”

More information on sources cited in-text

Citation Problems

- How can I keep track of all my PDFs and keep them organized?
- How can I find the right article to cite in my own work?
- Writing citations and bibliographies is tedious and boring

One Solution: Zotero



Zotero

Open-Source Citation Management Platform

https://guides.library.iit.edu/zotero

Galvin Library Zotero LibGuide

Metadata

- "Data whose purpose is to describe and give information about other data." - OED
  - Data about data

- Metadata about articles, books, etc:
  - Title
  - Author
  - Journal
  - Publication Year
  - Volume, Issue, Page Number
  - Keywords
  - Topics
  - Abstracts
  - Institutional Affiliations
  - Web Links
  - Unique Identifiers (ID Numbers)
  - DOIs or Persistent Links

- Metadata can be used to automatically build citations
  - Zotero is great for this

ZoteroBib

www.zbib.org

Simple citation construction using a web link

Builds citations using metadata in various citation styles

Zotero

Full Citation Management

www.zotero.org

Cloud version (Zotero Web Library)

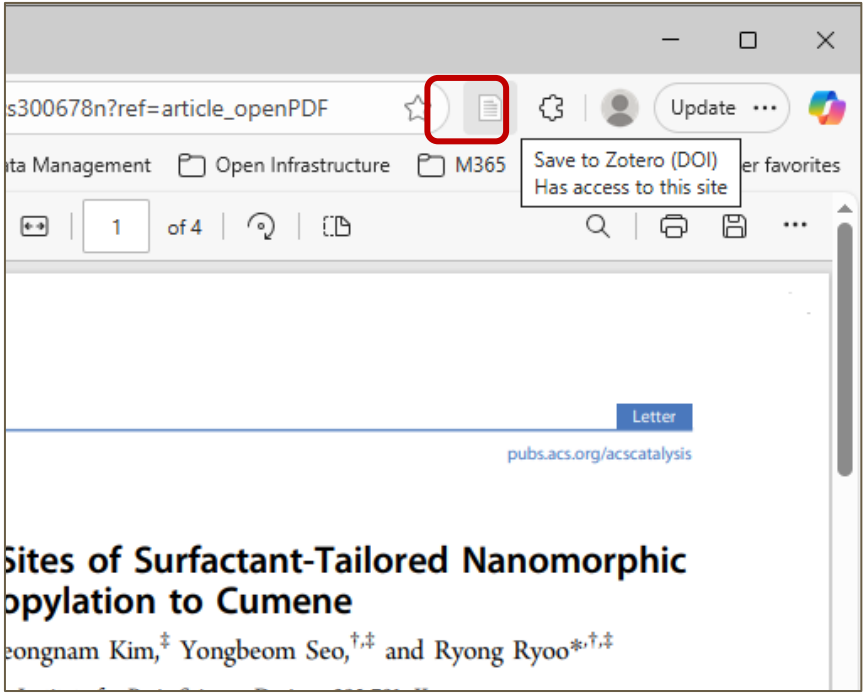
Standalone application

Browser Extension

Word Extension

Zotero Web Library

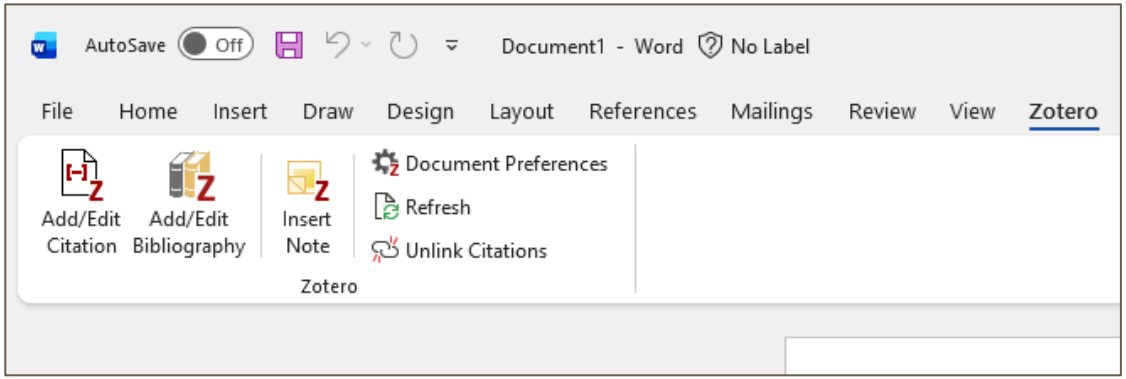
- Sign up for a free Zotero account
- Download Zotero Connector for your browser
  - Make the extension visible in your toolbar
- Find articles online
  - Galvin Library catalog
  - Google Scholar
- Save both PDFs and metadata to Zotero Web Library
  - 300 MB free space, Upgradeable at cost



Zotero Desktop

- Install on your computer
- Uses computer storage for PDFs, but can sync with Zotero Web Library
- More robust tools and options
- Required to use the Microsoft Word extension

(*Zotero | www.zotero.org*)



Zotero Plugin for Microsoft Word

- **Easily add citations from your Zotero library**
  - **In-Text**
  - **Bibliography**
- **Desktop version of Word Only**
- **Installs alongside Zotero Desktop**
- **Zotero Desktop and Word must be open at the same time**

Still can't find what you need?

The library can help!

- Research Help
  - Reference Desk
  - Research Appointments
- Interlibrary Loan
  - Worldwide Article Access

Sean Murphy

Head of Systems and Open Infrastructure

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Research Appointments

Main Library Desk